Simplifying Surds

Exam code: 0607

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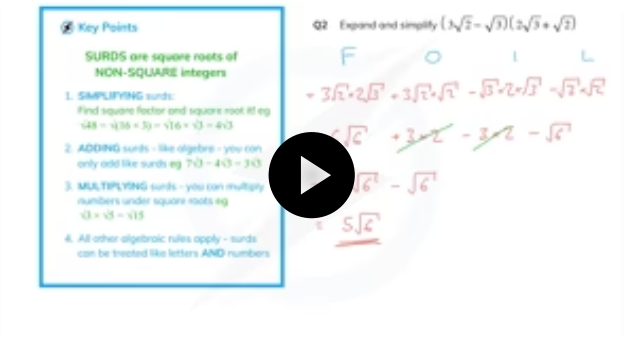

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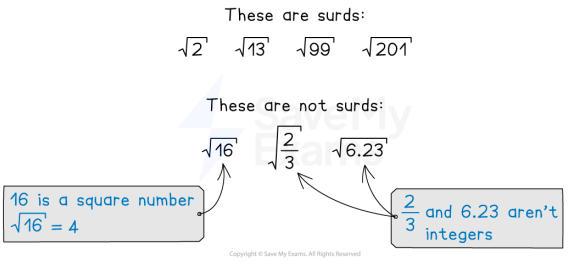
Surds & exact values

What is a surd?

- A surd is the square root of a non-square integer
- Using surds lets you leave answers in exact form
  - e.g. $5\sqrt{2}$ rather than $7.071067812...$



How do I do calculations with surds?

- **Multiplying surds**
  - You can multiply numbers under square roots together
  - $\sqrt{3} \times \sqrt{5} = \sqrt{3 \times 5} = \sqrt{15}$
- **Dividing surds**
  - You can divide numbers under square roots

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- You can factorise numbers under square roots
- $\sqrt{35} = \sqrt{5 \times 7} = \sqrt{5} \times \sqrt{7}$

- **Adding or subtracting surds**
  - You can only add or subtract multiples of "like" surds
    - This is similar to collecting like terms when simplifying algebra
  - $3\sqrt{5} + 8\sqrt{5} = 11\sqrt{5}$
  - $7\sqrt{3} - 4\sqrt{3} = 3\sqrt{3}$
    - However $2\sqrt{3} + 4\sqrt{6}$ cannot be simplified
  - You cannot add or subtract numbers under square roots
  - Consider $\sqrt{9} + \sqrt{4} = 3 + 2 = 5$
    - This is not equal to $\sqrt{9+4} = \sqrt{13} = 3.60555...$

Examiner Tips and Tricks

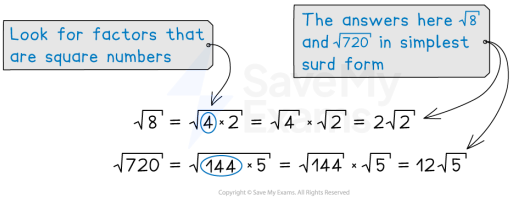
If your calculator gives an answer as a surd, leave the value as a surd throughout the rest of your working.

This will ensure you do not lose accuracy throughout your working.

Simplifying surds

How do I simplify surds?

- To **simplify a surd**, factorise the number using a square number, if possible
  - If multiple square numbers are a factor, use the largest
- Use the fact that $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$ and then work out any square roots of square numbers
  - E.g.
    $\sqrt{48} = \sqrt{16 \times 3} = \sqrt{16} \times \sqrt{3} = 4 \times \sqrt{3} = 4\sqrt{3}$



- When simplifying multiple surds, simplify **each separately**
  - This may produce surds which can then be collected together
    - E.g. $\sqrt{32} + \sqrt{8}$ can be rewritten as $\sqrt{16}\sqrt{2} + \sqrt{4}\sqrt{2}$

- You may have to expand double brackets containing surds
  - This can be done in the same way as multiplying out double brackets algebraically, and then simplifying
  - The property $(\sqrt{a})^2 = a$ can be used to simplify the expression, once expanded
  - E.g. $(\sqrt{6}-2)(\sqrt{6}+4)$ expands to $(\sqrt{6})^2 + 4\sqrt{6} - 2\sqrt{6} - 8$
    - This simplifies to $6 + 2\sqrt{6} - 8$ which gives $-2 + 2\sqrt{6}$

Worked Example

Write $\sqrt{54} - \sqrt{24}$ in the form $\sqrt{q}$ where q is a positive integer.

Answer:

Simplify both surds separately by finding the highest square number that is a factor of each of them

9 is a factor of 54, so $\sqrt{54} = \sqrt{9 \times 6} = 3\sqrt{6}$

4 is a factor of 24, so $\sqrt{24} = \sqrt{4 \times 6} = 2\sqrt{6}$

Simplify the whole expression by collecting the like terms

$$\sqrt{54} - \sqrt{24} = 3\sqrt{6} - 2\sqrt{6} = \sqrt{6}$$

$$\sqrt{6}$$

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